MIL-F-5591C
23 July 1986
SUPERSEDING
MIL-F-5591B
19 November 1963

MILITARY SPECIFICATION

FASTENERS; PANEL; NONSTRUCTURAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers nonstructural panel fasteners used on internal or external parts of aircraft.

1.2 Classification. Panel fasteners shall be of the following types, classes, styles, and sizes, as specified (see 3.2 and 6.2):

Type I	-	Flush head
Type II	-	Oval head
Type III	-	Wing head
Class A	-	Floating receptacle
Class B	-	Rigid receptacle
Style 1	-	Type I for deep dimple without grommet; types II and III for undimpled sheet without grommet.
Style 2	-	Type I for deep dimple with grommets; types II and III for undimpled sheet with grommets.
Style 3	-	Type I for shallow dimple with grommet; types II and III for undimpled sheet with grommet.
Sizes	-	Each head type shall be furnished in the sizes shown on figure 1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-A-250/4	Aluminum Alloy Plate and Sheet 2024
QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-H-1581	Hardware (Fasteners and Related Items) Packaging and Packing for Shipment and Storage of

MILITARY

DOD-D-1000	Drawings, Engineering and Associated Lists
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-18728	Steel Plate, Sheet, and Strip, Alloy 8630, Aircraft Quality
MIL-S-18729	Steel Plate, Sheet, and Strip, Alloy, 4130, Aircraft Quality

STANDARDS

MILITARY

MIL-STD-105	Sampling, Procedures and Tables for Inspection by Attributes
MIL-STD-1312	Fastener, Test Methods

(Copies of specifications, standards, drawings and publications required by suppliers in accordance with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The fasteners furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 Design. The style 1 fastener shall be of the stud-receptacle design. When the fastener is opened, the stud assembly shall be retained by the outer panel and the receptacle shall be retained by the panel support.

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The styles 2 and 3 fasteners shall be of the stud-grommet-receptacle design. When the fastener is opened, the stud shall be retained in the panel. The grommet shall be designed to guide the stud into the outer panel and to prevent contact between the outer panel and the stud and shall be independently retained in the outer panel.

The design of each style fastener shall permit the removal of the stud or stud-grommet combination and shall permit replacement with a fastener conforming to the dimensions specified in Table I.

Grommets shall be removable without damage to the panel.

3.2.1 Dimensions. Overall dimensions, rivet spacing, rivet hole size, and diameter of dimple and hole shall be as specified in Table I.

3.2.2 Engaging. The stud shall engage the receptacle within 180 degrees of rotation in the clockwise direction as viewed from the outside panel.

3.2.3 Locking. The stud shall be positively locked within 85 to 135 degrees of clockwise rotation, as viewed from the outside panel after the stud has been engaged.

3.2.3.1 Stop. A positive stop shall be provided to prevent the stud from being turned past the position of full lock.

3.2.4 Unlocking. The stud shall unlock within 85 to 135 degrees of counterclockwise rotation, as viewed from the outside panel.

3.2.5 Heads.

3.2.5.1 Flush head - type I. The stud heads shall be flush with the surface of the outer panel within + 0.015 when the fasteners are installed in mounting holes conforming to the dimensions specified in Table I and locking the total applicable panel thickness.

3.2.5.2 Oval head - type II. Application of the oval head studs shall not require dimpling or countersinking of the outer panel.

3.2.5.3 Wing head - type III. The wing head fastener shall be designed for operation without tools. Application shall not require dimpling or countersinking of the outer panel.

3.2.6 Screwdriver slots. The screwdriver slot in the head types I and II shall be such that a conventional blade-type screwdriver may be used to lock and unlock the fastener. When the fastener is in the locked position the slot shall be in line with the receptacle mounting rivets.

3.2.7 Floating receptacle. The class A receptacle shall be floating and shall provide a minimum lateral movement as specified in Table II. Lateral movement is measured in all directions from the center or neutral position.

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TABLE I. Panel fastener dimensions.

STYLE	CLASS	SIZE NUMBER	TYPE	A DIA MAX	B DIA MAX	D DIA MAX	D DIA MIN	F MAX	H DEGREES MAX	H DEGREES MIN	J DIA	K ±0.005	L MAX	H MAX	N MAX
1	A	7	I	0.919	0.390	0.720	0.700	0.774	120	118	0.125	1.375	0.625	2.000	1.156
			II, III	.919	.390	-	-	-	-	-	.125	1.375	.625	2.000	1.156
1	B	7	I	.882	.390	.720	.700	.774	120	118	.125	1.375	.625	2.000	1.000
			II, III	.822	.390	-	-	-	-	-	.125	1.375	.625	2.000	1.000
1	A	5	I	.807	.327	.650	.630	.713	120	118	.125	1.000	.562	1.625	1.032
			II, III	.807	.327	-	-	-	-	-	.125	1.000	.562	1.625	1.032
1	B	5	I	.713	.327	.650	.630	.713	120	118	.125	1.000	.562	1.625	0.875
			II, III	.713	.327	-	-	-	-	-	.125	1.000	.562	1.625	0.875
1	A	2	I	.557	.233	.416	.396	.478	100	98	.094	0.750	.438	1.375	0.598
			II, III	.557	.233	-	-	-	-	-	.094	0.750	.438	1.375	0.598
1	B	2	I	.510	.233	.416	.396	.478	100	98	.094	0.750	.438	1.375	0.562
			II, III	.510	.233	-	-	-	-	-	.094	0.750	.438	1.375	0.562
2	A	7	I	.945	.448	.832	.700	.882	120	118	.125	1.375	.625	2.000	1.156
			II, III	.902	.476	-	-	-	-	-	.125	1.375	.625	2.000	1.156
2	B	7	I	.902	.448	.832	.700	.882	120	118	.125	1.375	.625	2.000	1.000
			II, III	.902	.476	-	-	-	-	-	.125	1.375	.625	2.000	1.000
2	A	5	I	.832	.385	.762	.630	.812	120	118	.125	1.000	.562	1.625	1.032
			II, III	.832	.476	-	-	-	-	-	.125	1.000	.562	1.625	1.032
2	B	5	I	.832	.385	.762	.630	.812	120	118	.125	1.000	.562	1.625	0.875
			II, III	.832	.476	-	-	-	-	-	.125	1.000	.562	1.625	0.875
2	A	2	I	.570	.292	.494	.396	.550	100	98	.094	0.750	.438	1.375	0.598
			II, III	.570	.292	-	-	-	-	-	.094	0.750	.438	1.375	0.598
2	B	2	I	.570	.292	.494	.396	.550	100	98	.094	0.750	.438	1.375	0.562
			II, III	.570	.292	-	-	-	-	-	.094	0.750	.438	1.375	0.562
3	A	7L 1/	I	.833	.478	.680	.660	.687	141	139	.125	1.375	.734	1.750	1.156
			II, III	.833	.478	-	-	-	-	-	.125	1.375	.734	1.750	1.156
3	A	7S 1/	I	.713	.478	.680	.660	.687	141	139	.125	1.000	.734	1.625	1.032
			II, III	.713	.478	-	-	-	-	-	.125	1.000	.734	1.625	1.032
3	B	7	I	.713	.478	.680	.660	.687	141	119	.125	1.000	.734	1.625	0.875
			II, III	.713	.478	-	-	-	-	-	.125	1.000	.734	1.625	0.875
3	B	2	I	.500	.257	.450	.437	.480	101	99	.094	0.750	.703	1.000	0.562
			II, III	.500	.257	-	-	-	-	-	.094	0.750	.703	1.000	0.562

1/ L and S indicate a long (L) and short (S) rivet spacing. The same stud and grommet may be used for both the L and S receptacles.

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TABLE II. Mechanical properties.

2/ Size No.		2		5	7	
Style		1 & 2	3	1 & 2	1 & 2	3
Locking torque, minimum	lb. in.	3.5	1.0	5.0	7.5	5.0
Locking torque, maximum	lb. in.	8	11	12	18	20
Unlocking torque, minimum	lb. in.	3.5	1.0	5.0	7.5	7.5
Unlocking torque, maximum	lb. in.	10.4	12.5	15.6	23.4	25.5
Initial tension, minimum	lb.	15	8	35	50	35
Rated tensile load	lb.	200	200	500	700	700
Rated shear load	lb.	200	200	500	700	700
Locking stop strength, minimum	lb. in.	30	30	60	80	60
Vibration requirements:						
Frequency	cpm ±100	3600	3600	3600	3600	3600
Amplitude	inches	1/16	1/16	1/16	1/16	1/16
Load	pounds	75	75	175	225	225
Endurance, minimum	cycles	5000	5000	25,000	40,000	40,000
Lateral movement of class A, types I, II and III receptacles, minimum	inches	0.031	0.031	0.062	1/ 0.062	1/ 0.062

1/ When receptacle with 1.000-inch rivet spacing is used, lateral movement may be 0.031 inch minimum.
2/ See figure 1.

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3.3 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. The drawing number requirements of DOD-D-1000 shall govern documentation of and changes in the manufacturer's part numbers.

3.4 Finish. Aluminum-alloy parts shall be coated with an anodic film in accordance with MIL-A-8625. Fasteners manufactured from steel or copper alloy shall be cadmium plated in accordance with QQ-P-416, type II, class 2 (see 4.6.1).

3.5 Performance characteristics.

3.5.1 Initial tension. The initial tension (bearing pressure) between the panels shall not be less than the value specified in Table II (see 4.4.3.1).

3.5.2 Sheet separation. The sheet separation shall be not greater than 3/64 inch at the rated tensile load as specified in Table II.

3.5.3 Torque. The fasteners shall show no fracture or deformation when subjected to the torque tests specified in 4.4.3.5 thru 4.4.3.8.

3.5.3.1 Maximum torque (lock and unlock). The maximum torque required to lock and unlock the fastener shall be not greater than the maximum locking and unlocking torques specified in Table II (see 4.4.3.5).

3.5.3.2 Minimum torque (lock and unlock). The minimum torque required to lock and unlock the fastener shall not be less than the minimum locking and unlocking torques specified in Table II (see 4.4.3.6).

3.5.3.3 Absolute maximum torque (lock and unlock). The absolute maximum torque required to lock and unlock the fastener shall be not greater than 133 percent of the maximum locking and unlocking torque values specified in Table II (see 4.4.3.7).

3.5.3.4 Absolute minimum torque (lock and unlock). The absolute minimum torque required to lock and unlock the fastener shall not be less than 67 percent of the minimum locking and unlocking torque values specified in Table II (see 4.4.3.8).

3.5.4 Tension. When subjected to the rated tensile load specified in Table II, the fastener shall not develop a permanent set in excess of 0.008 inch for sizes 5 and 7, and 0.006 inch for size 2. There shall be no failure at a load less than 150 percent of the rated tensile load (see 4.4.3.10).

3.5.5 Shear. There shall be no permanent deformation when the fastener is subjected to the rated shear loads specified in Table II. There shall be no failures at less than 150 percent of the rated load (see 4.4.3.11).

3.5.5.1 The sheet thickness for the shear test specified in 4.4.3.11 shall be 0.064 inch for size 7 fasteners, 0.051 inch for size 5 fasteners and 0.040 inch for size 2 fasteners.

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3.5.6 Vibration. The fastener shall conform to the vibration requirements specified in Table II, without failure, when tested in accordance with 4.4.3.12.

3.5.7 Endurance. The fasteners shall withstand the number of cycles of operation specified in Table II without failure, when tested in accordance with 4.4.3.13.

3.5.8 Locking stop torque. The fastener, mounted as shown on figure 2, shall support without damage to the stud or receptacle, the minimum locking stop torque values specified in Table II (see 4.4.3.9).

3.5.9 Curved sheet installation. There shall be no interference with installation or removal when a rigid panel equipped with Style 1, 2 or 3 fasteners is in contact with a rigid substructure shown on figure 3 (see 4.4.3.14).

3.5.10 Corrosion resistance. There shall be no visible signs of corrosion of the fastener when tested as specified in 4.4.3.15.

3.5.11 Hardness. Panel fastener studs shall be hardened to prevent damage to such items as screwdriver slots. Studs shall be either carburized and surface hardened to a depth of 0.005 to 0.012 inch and to a superficial hardness of 78 to 90 on the Rockwell 15N scale, or, if fabricated of hardened steel, shall be heat treated to a Rockwell hardness of 30 to 41 HRC (see 4.4.3.16).

3.6 Failure. Failure as specified in 3.5.4, 3.5.5, 3.5.6, and 3.5.7 shall be considered as the point at which one or more of the fastener component parts fracture or break or where distortion and permanent set prevent the locking and unlocking of the fastener.

3.7 Workmanship. Fastener parts shall be free from burrs, tool marks, and other defects which may affect performance as defined by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality

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program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Inspection lot. An inspection lot shall consist of all fasteners of one type and size made from the same batch of material. The sample unit shall be one complete fastener assembly.

4.3 Classification of inspections. The inspection and testing of fasteners shall be classified as follows:

a. Qualification inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.4 Qualification inspections. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.4.1 Sample size. Samples for qualification shall consist of eight specimens as shown on figure 2, eight specimens as shown on figure 4, one specimen as shown on figure 4, mounted on sheets having a total thickness of 0.010 inch less than the total nominal thickness specified on figure 4, and one as shown on figure 3. Samples submitted for qualification testing shall be accompanied by three copies of test reports and two complete sets of manufacturing detail drawings of the sample submitted for test. After approval is granted, the design of the fastener shall not deviate from the samples and drawings submitted without authorization from the Government. Samples shall be forwarded to the activity responsible for qualification (see 6.3). The samples shall be plainly identified by securely attached durable tags, marked with the following information:

Sample for qualification test
FASTENERS; PANEL; NONSTRUCTURAL
Manufacturer's Part No.
Name of manufacturer
Submitted by (name) (date) for qualification test in accordance with requirements of MIL-F-5591C under authorization (reference authorizing letter)

4.4.2 Qualification test routine. The qualification tests shall be performed on the samples in the order listed below.

Four samples mounted in the fixture shown on figure 2 shall be subjected to the following tests:

a. Locking torque (see 4.4.3.3).

b. Unlocking torque (see 4.4.3.4).

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c. Initial tension (see 4.4.3.1).

d. Sheet separation (see 4.4.3.2).

e. Tension (see 4.4.3.10).

Four samples mounted in the fixture shown on figure 2 shall be subjected to the following tests:

a. Vibration (see 4.4.3.12).

b. Endurance (see 4.4.3.13).

Four samples mounted in the fixture shown on figure 4 shall be subjected to the following tests:

a. Locking torque (see 4.4.3.3).

b. Unlocking torque (see 4.4.3.4).

c. Absolute maximum and minimum locking and unlocking torque (see 4.4.3.7 and 4.4.3.8).

d. Shear (see 4.4.3.11).

Two samples mounted in the fixture shown on figure 4 shall be subjected to the corrosion-resistance test (see 4.4.3.15).

Two samples as shown on figure 4 shall be subjected to the hardness test (see 4.4.3.16).

Two samples mounted in the fixture shown on figure 4 shall be subjected to the locking stop torque test (see 4.4.3.9).

One sample mounted in the fixture shown on figure 3 shall be subjected to the curved-sheet installation test (see 4.4.3.14).

Two samples mounted in the fixture shown on figure 4 shall be modified and tested as specified in 4.4.3.8.

4.4.3 Qualification tests.

4.4.3.1 Initial tension. The initial tension shall be established by determining the tensile load that produces a separation of 0.003 inch of the area immediately around the fastener.

4.4.3.2 Sheet separation. The fasteners shall be loaded to the tensile value specified in Table II. The sheet separation of the area immediately around the fastener shall be measured.

4.4.3.3 Locking torque. The torque required to lock the fasteners shall be measured on unlubricated fasteners mounted in the fixture shown on figures 2 and 4 and as specified in 4.4.3.5 thru 4.4.3.8.

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4.4.3.4 Unlocking torque. The torque required to unlock the fastener shall be determined on the samples which have been subjected to the locking torque test (4.4.3.3).

4.4.3.5 Maximum torque (lock and unlock). The maximum torque test shall be conducted on test specimens fabricated to the nominal thickness of panels shown mounted in the fixture on figure 2.

4.4.3.6 Minimum torque (lock and unlock). The minimum torque test shall be conducted on test specimens fabricated to the nominal thickness of panels shown mounted in the fixture on figures 2 and 4.

4.4.3.7 Absolute maximum torque (lock and unlock). The absolute maximum torque test shall be conducted on test specimens with a 0.010 inch shim between the panels shown mounted in the fixture on figures 2 and 4.

4.4.3.8 Absolute minimum torque (lock and unlock). The absolute minimum torque test shall be conducted on test specimens having a total thickness of 0.010 inch less than the total nominal thickness shown on figure 2.

4.4.3.9 Locking stop torque. The locking stop torque specified in Table II shall be applied to the fastener through a screwdriver bit.

4.4.3.10 Tension. Upon completion of the locking and unlocking torque tests, the rated tensile load specified in Table II shall be applied normal to the test sheet five consecutive times without failure in accordance with MIL-STD-1312, Test 23. The fastener shall then be subjected to the locking and unlocking torque tests. The fastener shall then be loaded to 150 percent of the rated tensile load specified in Table II.

4.4.3.11 Shear. The fastener shall be tested as specified in MIL-STD-1312, Test 4, alternate test specimen configuration, except that the mounting shall conform to the requirements of figure 4. The fastener shall then be subjected to the locking and unlocking torque tests. The fastener shall then be loaded to 150 percent of rated shear load. There shall be no failure.

4.4.3.12 Vibration. The fastener shall be mounted and loaded as shown on figure 5. The fastener shall be vibrated with a simple harmonic motion of 1 degree of freedom for 25 hours in conformance to 3.6.6. The load, amplitude, and frequency shall be as specified in Table II. The fastener shall then be vibrated without load for an additional 5 hours without failure.

4.4.3.13 Endurance. The fasteners shall be subjected to the number of cycles of opening and closing operations specified in Table II at 110 cpm. The test shall be performed on fasteners which have passed the vibration test (4.4.3.12). A drop of oil may be added at the beginning of the test.

4.4.3.14 Curved-sheet installation. The fasteners shall be installed in accordance with figure 3. They shall then be operated to determine if there is interference with installation, sheet separation, or removal. In the event that interference is experienced with the style 1 fasteners, studs shall be

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progressively removed from one end of the specimen until the outer panel can be installed or removed without interference. The included angle of the fasteners remaining after such removal shall be recorded.

4.4.3.15 Corrosion resistance. The fastener shall be subjected to salt spray test in accordance with MIL-STD-1312, Test 1, except that test duration shall be 200 hours. The specimen shall be in accordance with figure 4 and shall not have been employed in other tests.

4.4.3.16 Hardness. The studs shall be tested in accordance with MIL-STD-1312, Test 6, except that the hardness of the studs shall be tested on the face of the stud on either side of the screwdriver slot. Case hardened surfaces shall be tested on the 15N Rockwell superficial scale. The heat treated, through-hardened studs shall be tested on the Rockwell C scale.

4.5 Quality conformance inspections. The quality conformance inspections shall consist of the tests and examinations specified in 4.5.2 and 4.6.

4.5.1 Sampling.

4.5.1.1 Sampling for visual examination. Random samples shall be selected from each lot in accordance with MIL-STD-105, inspection level II. The acceptable quality level (AQL) shall be 2.5 percent defective (see 4.6.1).

4.5.1.2 Sampling for tests. From each lot (see 4.2), a random sample shall be selected in conformance to MIL-STD-105, inspection level S-2. The AQL shall be 1.0 percent defective.

4.5.2 Quality conformance tests. Each sample fastener shall be subjected to the following tests:

a. Torque (see 4.4.3.3 thru 4.4.3.8).

b. Locking stop torque (see 4.4.3.9).

c. Hardness (see 4.4.3.16).

4.6 Examinations.

4.6.1 Examination of product. Each fastener shall be visually examined for finish (see 3.4) and workmanship (see 3.7).

4.7 Inspection for preservation, packaging and delivery. Preparation for delivery shall be inspected for conformance to Section 5.

5. PACKAGING

5.1 Preservation - packaging.

5.1.1 Levels A and C. The fastener shall be preserved and packaged in accordance with PPP-H-1581.

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5.2 Packing.

5.2.1 Levels A, B and C. The fastener shall be packed in accordance with PPP-H-1581. Unless otherwise specified by the procuring activity, quantities per unit package shall be in accordance with PPP-H-1581.

5.3 Marking. Marking shall be in accordance with PPP-H-1581.

6. NOTES

6.1 Intended use. The panel fasteners covered by this specification are for use on aircraft, where a readily operated fastener is required, such as panels, inspection doors, and quick detachable plates which do not carry primary structural stresses.

Type I flush-head fasteners are intended for use on the external surfaces of aircraft. Types II and III fasteners are intended for interior use and where the fastener stud heads are not subject to airflow.

Styles 1 and 2, Type I, fasteners are suitable for use in the more ductile materials where dimpling for styles 1 and 2 may be accomplished without heated dimpling equipment. They are also suitable for use in the harder materials where heated dimpling equipment must be used in forming the type I dimples.

Style 3, Type I, fasteners are suitable for installation in aluminum alloys, magnesium alloys, and corrosion-resistant steel without the use of heated dimpling equipment in forming the dimples for Type I fasteners.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Type, class, style and size (see 1.2).

c. Levels of packaging and packing (see 5.1 and 5.2).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to the requirements, and manufacturers are urged to arrange to have the products that they propose to offer the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Navy Department, Washington, DC 20460. However, information pertaining to qualification of products may be obtained from the Naval Air Development Center, Code 6013, Warminster, PA 18974.

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